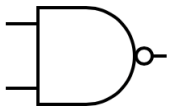
1. What is the name of this gate?



   (a) AND.
   (b) NAND.
   (c) NOT.
   (d) NOR.
   (e) XOR.

2. Which of these equations would produce the following truth table?

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

   (a) $Q = \bar{A} + \bar{B} + A + \bar{B}$.
   (b) $Q = \bar{A} + A \cdot \bar{B} + B$.
   (c) $Q = \bar{A} + \bar{B} \cdot A + \bar{B}$.
   (d) $Q = \bar{A} \cdot B + A \cdot \bar{B}$.
   (e) $Q = \bar{A} \cdot \bar{B} \cdot A \cdot \bar{B}$.

3. How many address lines are necessary to access 128KB of memory

   (a) 8.
   (b) 11.
   (c) 15.
   (d) 16.
   (e) 17.

4. Which addressing mode is used in the 6809 instruction `LDA #$F0`?

   (a) Immediate.
   (b) Indirect.
   (c) Indexed.
   (d) Indexed Indirect.
   (e) Extended.

5. Which flag on the 6809 stores whether or not the result of the last instruction was negative?

   (a) F.
   (b) V.
   (c) Z.
   (d) N.
   (e) C.

6. In C, which of the following allocates 8 bytes of memory on the stack?

   (a) `uint32_t z;`
   (b) `int8_t[] y[4];`
   (c) `uint8_t z;`
   (d) `uint8_t x[8];`
   (e) `uint16_t z;`

7. In C, which of the following declares a pointer to a routine that takes an integer parameter and returns a pointer to a double:

   (a) `double *method(int param);`
   (b) `double method(int param);`
   (c) `double *(*method)(int param);`
   (d) `double (*method)(int param);`
   (e) `double (*method)(int *param);`

8. Given the following page table, and a page size of 4, which one of the following physical addresses is mapped from the logical address 1001?

page	frame
00	110
01	100
10	011
11	101

   (a) 10001.

   (b) 11001.

   (c) 01101.

   (d) 01110.

   (e) 1001.

9. Consider the following sequence of references to page IDs caused by a process: 6, 7, 8, 9, 7, 6, 10, 11, 7, 6, 7, 8, 12, 11, 8, 7, 6, 7, 8, 11. Assume the process has only four physical frames and all entries of the page table are initially empty. How many page faults would result from this sequence of references using the least-recently-used (LRU) page replacement algorithm?

   (a) 8.

   (b) 9.

   (c) 10.

   (d) 11.

   (e) None of the above.

10. The following table gives information about 4 processes: their IDs, their arrival times, their initial CPU burst times (given in milliseconds), and their priorities (low numbers denote high priorities). Suppose *Round-Robin (RR)* scheduling with a *time quantum of 3 ms* is used, which one of the following scheduling orders is correct? Note that, when all the criteria of the scheduling algorithm are equal for two processes, order them by process ID (1–4).

Process ID	Arrival Time (ms)	CPU Burst Time (ms)	Priority
1	1	8	2
2	5	4	1
3	1	7	4
4	2	3	3

(a) P1—P3—P4—P1—P2—P3—P1—P2—P3.

(b) P1—P2—P3—P4—P1—P2—P3—P4.

(c) P1—P2—P3—P4.

(d) P1—P4—P3—P2—P1—P4—P3—P2.

(e) P1—P3—P4—P1—P2—P3—P1—P2.

11. The following table gives information about 4 processes: their IDs, their arrival times, their initial CPU burst times (given in milliseconds), and their priorities (low numbers denote high priorities). Suppose *First-Come-First-Served (FCFS)* scheduling is used. What is the average wait time for the processes? Note that, when all the criteria of the scheduling algorithm are equal for two processes, order them by process ID (1–4).

Process ID	Arrival Time (ms)	CPU Burst Time (ms)	Priority
1	1	8	2
2	5	4	1
3	1	7	4
4	2	3	3

(a) 8.1 ms

(b) 10 ms

(c) 11 ms

(d) 16 ms

(e) 9 ms

END